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Design Expanded BCI with Improved Efficiency for VR-embedded NeuroRehabilitation Systems

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Abstract—**A general brain computer interface (BCI) usually consists of three main units known as preprocessing unit, feature selection unit and classification unit. In this paper, an EEG-based BCI with expanded structure is introduced that provides opportunity to improve efficiency of virtual reality (VR) embedded neurorehabilitation systems. The proposed BCI has to detect three different neuro-stimulations during specified motor imagery tasks and generate proper virtual neuro-stimulations for the avatar to do the task in the VR world. In the proposed BCI, discrete wavelet transformation (DWT) and multilayer perceptron (MLP) neural network are applied for preprocessing and classification, respectively; and an expounder is added to eliminate misclassifications which lead to wrong virtual neuro-stimulations. Offline EEG signals are applied to examine the proposed BCI and results are demonstrated.**

Keywords- brain computer interface(BCI); EEG motor imagery signals; virtual reality; neuro-rehabilitation

I. INTRODUCTION

Motor function degeneration is a common result in many stroke patients that causes upper or lower limb impairment. Assistive devices and rehabilitation systems are developed to improve patient's abilities. In the past years, various types of rehabilitation robots are introduced [1]. Rehabilitation robots come in close contact with impaired limb and make mutual physical interaction with patient [2]. Ensuring safe and effective physical interaction implies critical requirements for design and control of Assistive devices [3]. As a result, Commercial rehabilitation devices are usually large and very expensive. Although, effectiveness of robotic rehabilitation methods is proven [4]; they have a deficiency when continuous passive motion (CPM) is exerted on severely disabled patients during long term rehabilitation treatments. In fact, applying CPM method without any brain excitation will result in neuro-motor function degeneration since patients cannot interact with robot cooperatively [5].

Brain computer interface (BCI) gives a solution to this problem. BCI make this possible to detect different neuro-stimulations and generate proper neuro-feedbacks. Up to now, different mental strategies are exploited by BCI systems such as P300 component analysis [6], visual evoked potentials, and slow cortical potentials and also the well-known motor imagery (MI). MI-based BCI typically is used in applications including smart orthoses and prostheses, wheelchair control and cursor control [7]. In the past decade, MI-based BCI has indicated useful applications in the neurorehabilitation field [8]. Several studies claimed that MI-based BCI can excite brain plasticity [9, 10]. Effectiveness of BCI-aided neurorehabilitation is assessed frequently and shows significant progress in neuro-motor recovery [11]. Recently, MI-based BCI along with virtual reality (VR) has introduced low priced, diverse and attractive types of rehabilitation systems called VR-embedded neurorehabilitation systems. VR provides an interesting game-like protocol for BCI-aided neurorehabilitation. VR makes this possible for patients to do the activities which they cannot do them in the real world through imagination. Also VR gives a chance to design motor imagery tasks based on daily living activities (DLA) and patient's priority. The main purpose of using VR in the neurorehabilitation systems as a visual feedback tool is to excite patient's brain to generate correct stimulations based on instructed motor imagery task and always encourage them with positive feedback which shows their mental ability to move.

To the best of author's knowledge, there is a few works among numerous researches that have employed VR as a visual feedback tool in the MI-based BCI-aided neurorehabilitation systems [12, 13]. In the most promised research in the field of human fist movements, it is shown that wrong visual feedbacks which are generated from misclassifications truly have affected neurorehabilitation efficiency by disturbing patients and disrupting their concentration and confidence [13].

Effective solution to the addressed problem is to develop more accurate classifiers and design an expounder unit to eliminate misclassifications which are studied in the current work. The paper is organized as follows: section II describes structure of the considered VR-embedded neurorehabilitation system and EEG signal data set. Section III introduces expanded structure of EEG-based BCI while Performance of the proposed BCI is examined in Section IV. Conclusion and future works are discussed in section V.

II. METHOD AND MATERIALS

A. VR-Embedded NeuroRehabilitation System

Structure of the considered VR-embedded neurorehabilitation system for human fists movement is shown in Fig. 1. In this architecture Real world is connected to an immersive virtual world through BCI instrument.

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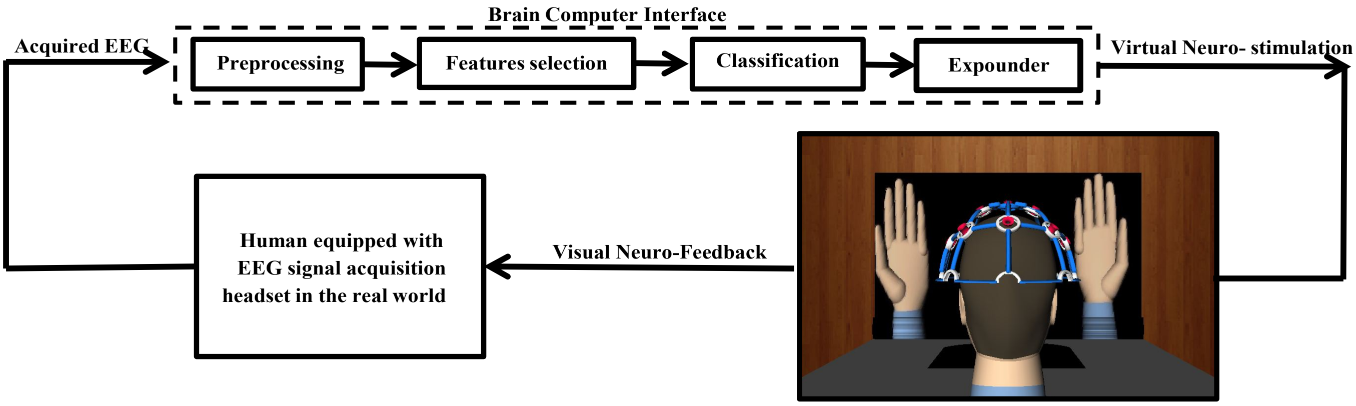


Figure 1. Structure of the considered VR-embedded neurorehabilitation system for human fists movement

BCI transports human into an avatar in the virtual world as it is depicted in Fig. 2 where the avatar is triggered to do the detected activities of human's brain. In the considered neurorehabilitation system, electroencephalography (EEG) signal is acquired from patient's brain while patient is requested to do cue-based motor imagery task. The instructed task is the act of opening/closing of right and left fists. The proposed BCI has to detect three different neuro-stimulations and generate proper virtual neuro-stimulations for the avatar to do the task in the VR world. Neural-stimulations come in three categories including right fist movement (RFM), rest mode (RM) and left fist movement (LFM) which are shown in Fig. 3. Although authors have already developed the immersive virtual reality environment so providing visual feedback is possible for online examinations; current research concentrates on design and development of the expanded BCI through offline EEG motor imagery signals.

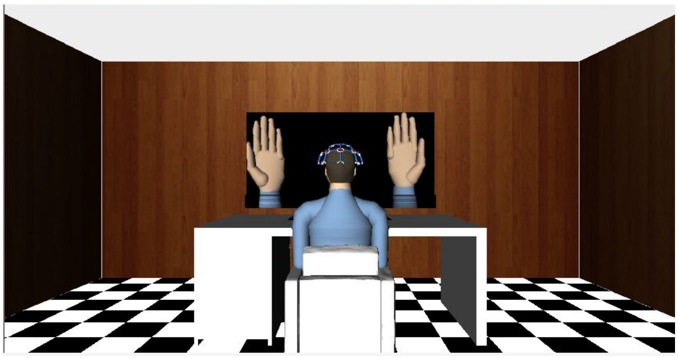


Figure 2. Immersive virtual reality environment developed for VR-embedded neurorehabilitation of human fists movement

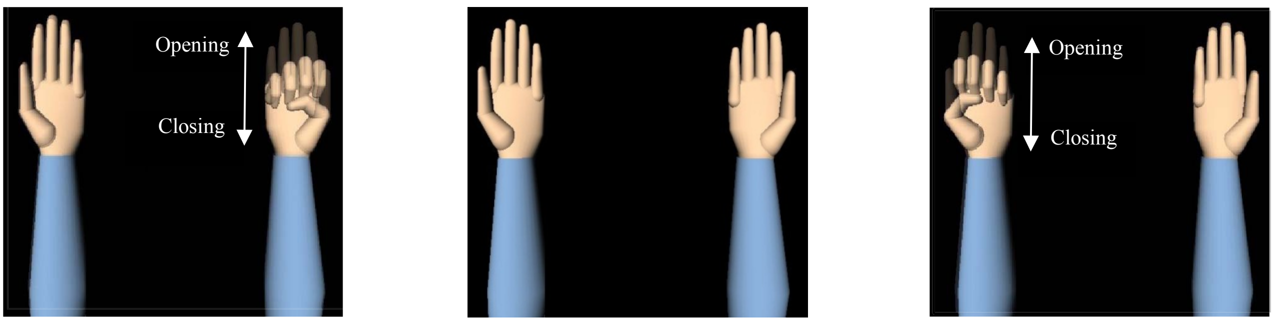
B. Data Base

EEG motor movement/imagery signals are obtained from PhysioNet [14] dataset which is created by BCI2000 instrumentation system [15]. Although, the dataset is collected from 109 volunteers; here data of first three subjects is used for off-line BCI application. As it is demonstrated in Fig. 4, EEG signals are recorded in the international 10-10 system with 64-channels. Also, motor cortex electrodes are highlighted in Fig. 4. The C_3, C_Z and C_4 channels are preferred to detect motor activities during RFM, RM and LFM, respectively. Each subject performs two specific motor task based on following instructions [14]:

Task 1: A target appears on either the left or the right side of the screen. The subject opens and closes the corresponding fist until the target disappears. Then the subject relaxes.

Task 2: A target appears on either the left or the right side of the screen. The subject imagines opening and closing the corresponding fist until the target disappears. Then the subject relaxes.

Task1 and Task 2 are repeated three times for subjects. Each task is executed in 2 minutes (120 sec) recording run and contains 30 four-second sequences. All of sequences comprise an annotation code to distinguish between RFM, RM and LFM by T_2, T_0 and T_1 codes, respectively. Consequently, Task1 and Task 2 distinguish between movement (Mov.) and imagery (Img.) tasks while annotation codes distinguish between neuro-stimulation categories. So the overall data base contains 540 different sequences. The data set is informed based on subject's contribution and motion categories in Table I and Table II, respectively.



(a) RFM (b) RM (C) LFM

Figure 3. Three kind of virtual Neural Stimulations: a) right fist movement (RFM), b) rest mode (RM) and c) left fist movement (LFM)

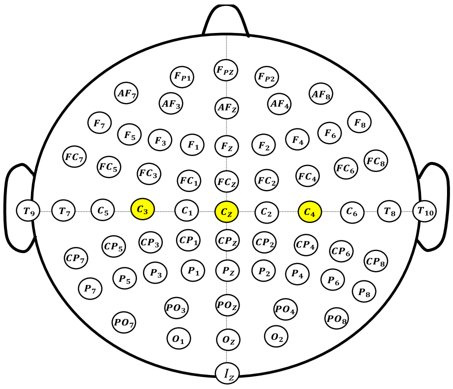


Figure 4. 64-Channel EEG electrodes in the international 10-10 system excluding N_Z, F_9, F_{10}, F_{T9}, F_{T10}, A_1, A_2, P_9, P_{10}, TP_9 and TP_{10} channels [15].

TABLE I. EEG signal dataset based on subject's contribution

subject	T_2 Mov.	T_2 Img.	T_0 Mov.	T_0 Img.	T_1 Mov.	T_1 Img.
S1	22	22	45	45	23	23
S2	22	22	45	45	23	23
S3	23	22	45	45	22	23

Total contribution of each subject is 180 sequences.

TABLE II. EEG signal dataset based on each motion category contribution

Motion Category	Movement	Imagery	Total
RFM	67	66	133
RM	135	135	270
LFM	68	69	137

Data base is constructed by total number of 540 sequences.

III. Brain Computer Interface

Brain computer interface translates the acquired EEG signals to the virtual neuro-stimulations by means of signal processing. BCI performs the signal processing task in four sequential steps containing preprocessing, feature selection, classification and interpretation.

A. Preprocessing

In the preprocessing step, all of the sequences should be denoised to eliminate effect of higher frequencies, baseline and EOG artifact. First, a 1^{th}-order butterworth high-pass filter at frequency of 0.5 Hz is applied. Then, an 8^{th}-order elliptic low-pass filter is used at frequency of 50 Hz. Therefore, a band-pass filter is obtained which passes through 0.5-50Hz frequency band. Performance of proposed band-pass filter is illustrated in Fig. 5, comparatively.

Hereafter, filtered sequences should be divided into frequency sub-bands which within α-band (8-12 Hz), β-band (12-30 Hz) and μ (< 10 Hz) rhythm are appeared. Wavelet transformation (WT) is a powerful time-frequency domain for signal analysis. WT widely is used in signal decomposition, de-noising, reconstruction and even feature extraction. Discrete wavelet transformation (DWT) decomposes the signal into approximate and detail components through (1) [16]:

$$DWT\ \phi(j,k) = \frac{1}{\sqrt{M}} \sum_{n=1}^{M-1} X(n)\phi jk(n) \tag{1}$$

where, ϕ is the mother wavelet, *X(n)* is the sampled signal, *j* is the scale parameter and translation parameter is *k*. Parameter *M* denotes signal length in samples. All of four-second sequences contain 640 samples due to sampling frequency of 160 Hz and its corresponding sampling interval of 0.00625 seconds. In the current study, 1-dimentional (1-D) discrete wavelet transformation is applied using wave menu graphical user interface (GUI) of MATLAB software. Signal decomposition is executed based on daubechies 2 wavelet transformation in four levels. Fig. 6 shows the result of wavelet transformation for T_2-type filtered sequence of Fig. 5.

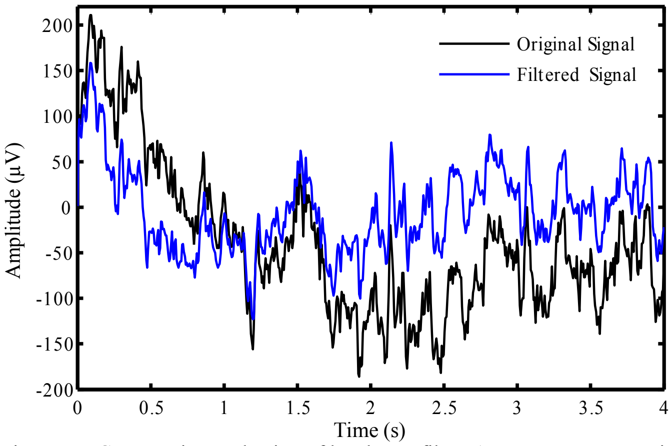
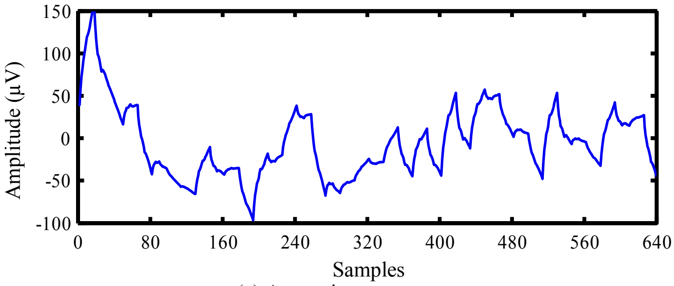
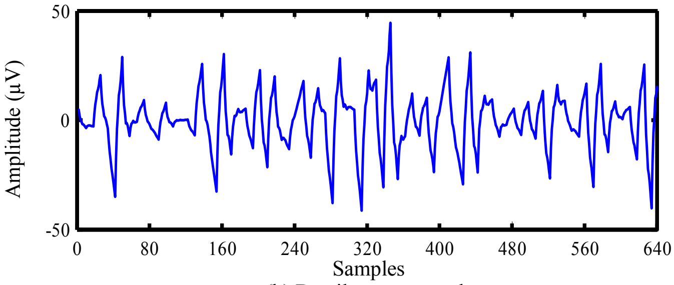


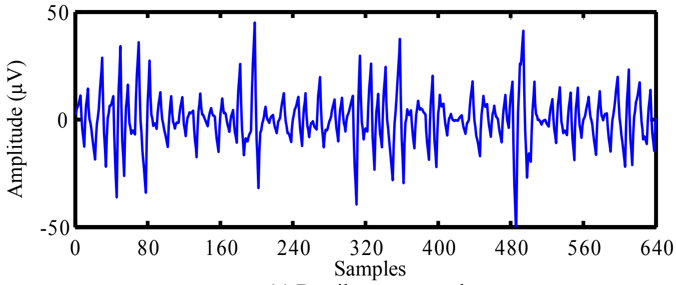
Figure 5. Comparative evaluation of band-pass filter. A T_2-taype sequence is extracted from C_3 channel during subject3's EEG motor movement task. Filtering response shows obvious decrease in DC-level and significant high frequency attenuation which is sensible in large scale illustration.



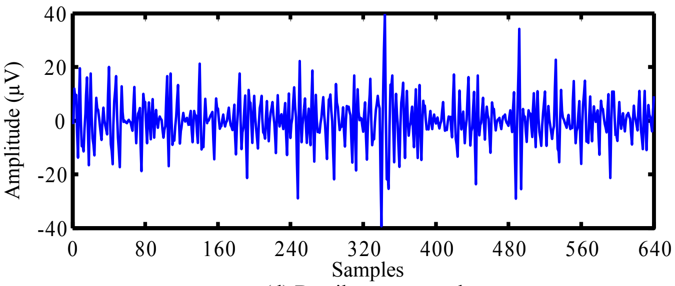
(a) Approximate component a_4



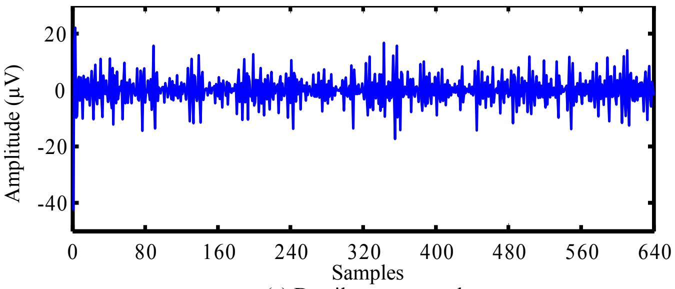
(b) Detail component d_4



(c) Detail component d_3



(d) Detail component d_2



(e) Detail component d_1

Figure 6. wavelet transformation result for T_2-type filtered sequence of Fig. 5. (a) Approximate components a_4 (b) Detail components d_4 (c) Detail components d_3 (d) Detail components d_2 (e) Detail components d_1

The approximate coefficient (a_4) contains the lowest frequency components of original signal. Approximate coefficient is known as the main part of the decomposed signal. Detail coefficients comprise higher frequencies such that the first detail coefficient has the highest frequency components. According to (2), original signal (*X*) can be reconstructed by its decompositions at level of *N*.

$$X = a_N + \sum_{n=1}^{N} d_n \tag{2}$$

B. Feature Selection

The approximate and the detail decompositions are used for feature extraction. Energy is considered as the main property of a signal to extract from decompositions. According to (3), energy is obtained for a discrete signal *X*:

$$E(X) = \sum_{n=1}^{M} |X[n]|^2 \tag{3}$$

where, *M* denotes signal length in samples. Feature vector is calculated for each sequence through (4). As it is demonstrated in Table III, a 5×540 feature matrix is obtained for 540 four-second sequences within feature vectors are columns of the feature matrix.

$$FV(seq) = [E(a_4)\, E(d_4)\, E(d_3)\, E(d_2)\, E(d_1)]^T \tag{4}$$

TABLE III. Structure of 5×540 feature matrix

Energy	T_2			T_0			T_1		
	Seq. 1	…	Seq. 133	Seq. 134	…	Seq. 403	Seq. 404	…	Seq. 540
$E(a_4)$	58.49	…	45.03	63.11	…	59.89	61.52	…	56.17
$E(d_4)$	1.665	…	6.617	1.798	…	4.66	1.817	…	3.623
$E(d_3)$	7.329	…	19.36	8.168	…	10.60	7.427	…	9.477
$E(d_2)$	13.29	…	12.73	12.74	…	10.56	11.70	…	12.20
$E(d_1)$	19.21	…	16.24	14.17	…	14.26	17.52	…	18.51

C. Classification

The multilayer perceptron (MLP) artificial neural network (ANN) is used for EEG signal classification. MLP neural networks work well with normalized data inputs. Consequently, the input features are normalized by (5) over the three classes (T_2, T_0 and T_1) separately.

$$\hat{f}(i,j) = \frac{f(i,j) - min_j}{max_j - min_j} \tag{5}$$

Symptoms min_j and max_j indicate minimum and maximum of the features matrix rows, respectively. The classifier will be trained based on the scaled conjugate gradient back-propagation learning function. Performance of the classifier is evaluated by mean squared error (MSE) function which is represented in (6):

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i)^2 \tag{6}$$

where, parameters $\hat{y}_i$ and y_i are classifier output and classification target, respectively. Parameter *n* denotes number of input samples. The MLP classifier is constructed by pattern recognition tools of MATLAB software. As it is demonstrated in Fig. 7, architecture of the MLP classifier is a two-layer feed-forward network with sigmoid activation functions. The classifier network is designed with 10 neurons in the hidden layer and 3 neurons in the output layer. The MLP classifier is trained by 45% of normalized database while it's 20% is used for classifier validation. The other 35% is used for classifier testing. Table IV determines contribution of each class and defines its corresponding target vector.

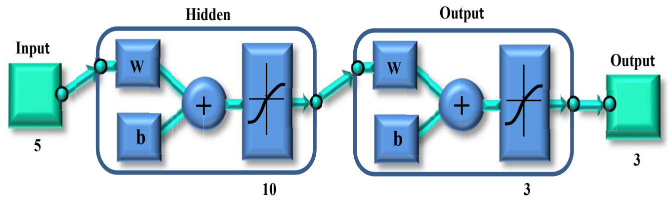


Figure 7. Architecture of the MLP classifier

TABLE IV. The MLP classifier metrics in training, validation and testing

Class	Input			Target
	Training	Validation	Testing	
T_2	60	22	51	$[1\ 0\ 0]^T$
T_0	125	61	84	$[0\ 1\ 0]^T$
T_1	58	25	54	$[0\ 0\ 1]^T$

D. Interpretation

In order to neglect misclassifications, an interpretation unit is considered. The interpretation unit which is called the expounder has to detect MLP classifier misclassifications and eliminate them from triggered virtual neuro-stimulations. The expounder detects misclassifications through a logical procedure which is described in (7):

$$\begin{Bmatrix} \text{round}(\hat{y}_i) \ \&\& \ y_i = y_i : \ classification \\ \text{round}(\hat{y}_i) \ \&\& \ y_i \neq y_i : misclassification \end{Bmatrix} \quad (7)$$

where parameters $\hat{y}_i$ and y_i are classifier output and classification target, respectively. Note that the procedure is only applicable to cue-based motor imagery tasks because, cue-based MI task is executed based on instructed scenario which is described by annotation codes. The annotation codes define target class (y_i) in each sequence of the task.

IV. PERFORMANCE EVALUATION

Confusion matrixes in Fig. 8 illustrate performance of the MLP classifier after 99 iterations. Fig. 8 indicates that the MLP classifier successfully recognizes 93.2% of T_2 class, 100% of T_0 class and 97.8% of T_1 class. Also total accuracy of the MLP classifier is 97.8%. The classifier failed in classification for 12 input samples which leads to 2.2% misclassification rate. 9 of the misclassifications belong to the T_2 class and 3 of them belong to the T_1 class.

Since misclassifications are occurred only in movement (class 1: RFM (T_2) and class 3: LFM (T_1)) classes, so the expounder eliminates them by generating null virtual stimulation as in RM. Consequently, during misclassifications patients are provided with RM visual feedback instead of viewing wrong hand movements. Throughout the null stimulations, patients have opportunity to rest and concentrate on the upcoming instructions. While without the null stimulation, concentration and confidence of patients will be negatively affected and reduces efficiency of the VR-embedded neuro rehabilitation system in neuro motor function recovery.

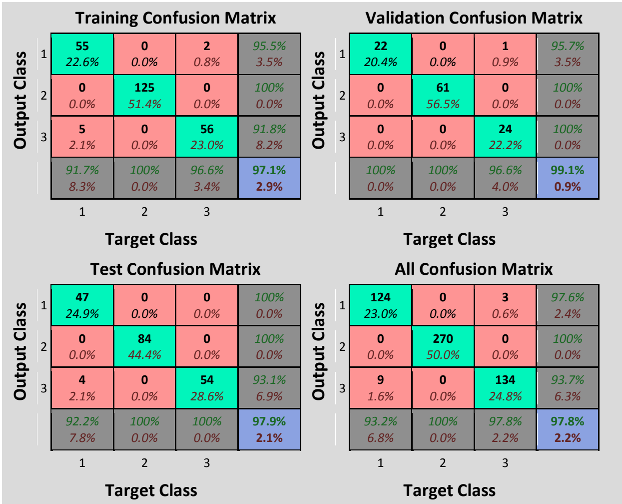


Figure 8. The MLP classifier Performance

In the following, the proposed expanded EEG-based BCI is examined by a pre-recorded motor imagery task which contains three misclassifications (i.e. record number 4 of the third subject). Simulation result of the investigated record is demonstrated in Fig. 9. Throughout the sequence number 2 (2-8 seconds), sequence number 10 (36-40 seconds) and sequence number 16 (60-64 seconds), the expounder has transformed misclassifications into the RM. The transformations are notified by downward arrows. In the rest of the sequences, since the MLP classifier has successfully recognized neuro-stimulations there is no transformation.

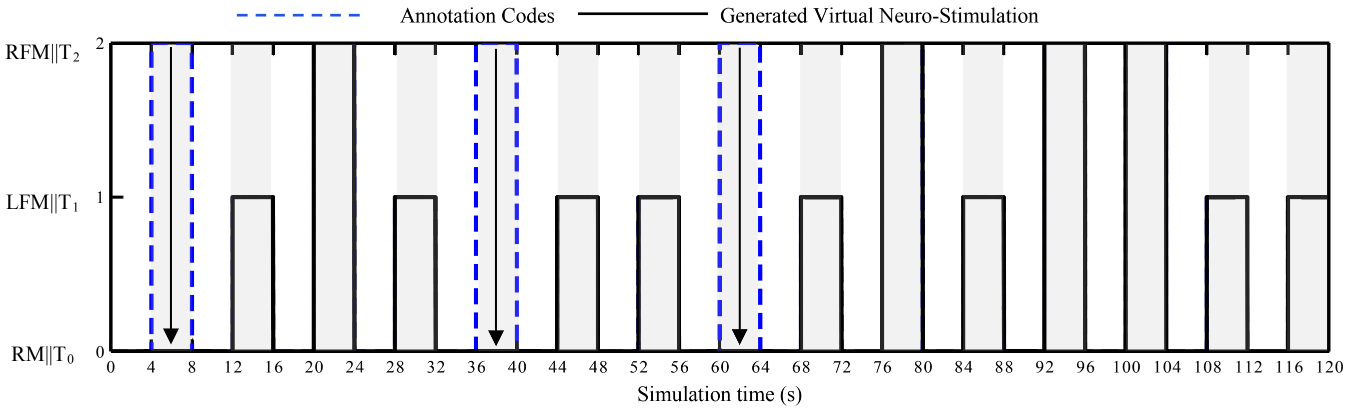


Figure 9. Examination of the proposed expanded BCI with a whole off-line record (i.e. record number 4 of the third subject).

Eventually, efficiency of the proposed expanded BCI for employed offline EEG motor imagery data base is reported in Table V. results indicates that 100% rate is achieved for positive virtual neuro-stimulation using the proposed expanded BCI that ensures always positive encourage for patients.

TABLE V. Efficiency of the proposed expanded BCI

Neuro-Stimulation	Recognition (Sample - Rate)	Transformation (Sample - Rate)	Overall Positive Virtual Stimulation (Sample - Rate)
RFM	124 - 93.2%	9 - 1.6%	124 - 100%
RM	270 - 100%	0 - 0%	282 - 100%
LFM	134 - 97.8%	3 - 0.6%	134 - 100%

Total recognition rate is 97.8% and total transformation rate is 2.2%.

V. CONCLUSION AND FUTURE WORK

Current research contributes to design and develop an EEG-based BCI with expanded structure. The proposed EEG-based BCI contains the following four units; a preprocessing unit which employs a band-pass filter and discreet wavelet transformation to prepare sequenced EEG signals for feature extraction; a feature selection unit which calculates energy of the sequenced EEG signals from its decompositions and forms the normalized feature vectors for classification; a classifier unit which is realized by multilayer perceptron (MLP) neural networks to detect three different neuro-stimulations; and a logical expounder unit to generate proper virtual neuro-stimulations from correct classifications and eliminate misclassifications by transforming them into the rest mode. The expanded BCI is designed such that ensures always positive encourage for patients and gives a chance to improve efficiency of the VR-embedded neurorehabilitation system.

Examination of the considered VR-embedded neurorehabilitation system with online EEG signals is the most preferred goal for the future work. Since recognition rate is significantly degraded in on-line BCI applications; obtaining better recognition rate for classifier over a bigger data base is already followed.

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